War and Peace: A Source-Generated Terminal Theorem — Technical Supplement

Technical Appendix: Exact Settlement, Open Revision, and Original History

This appendix gives the definitions, decisive arguments, and source correspondence for PE1–PE26; execution states, stocks, physical clocks, and original debt keys are written here as x, s_x, t_x, a (corresponding to s, σ, τ, ρ in the main text). Certified histories and handoff dispositions have a separate execution appendix. Each material good retains its own unit; reparations, financial principal, food, and review charges have separate accounts.

A. Original histories, losses, the unilateral bound, and the viable settlement domain

Let participants, goods, ports, technologies, financial instruments, and communities range over given types. An actual state x contains stocks $s_x(p, g) \in \mathbb{Q}$, a physical clock t_x, the priority pending queue, the visit ordinal $\mathrm{ord}(x) \in \mathbb{N}$, the operative technology, and the complete responsibility ledger. The institution supplies authorization, actual debits and credits, and the successor of a literal action. A submission can emit several material items. Visit ordinal, material-item ordinal, and physical time are recorded separately.

A positive net debit across a community boundary, made without the affected person's consent, is addressed by

$$a = (\text{submission ordinal}, \text{actual-item ordinal}, (\text{victim}, \text{payer}, \text{good})).$$

The original loss at a is $d_a > 0$, and its outstanding balance is $b_a \geq 0$. Subsequent financial registration and institutional revision retain the original provenance of this address. If a submission is rejected because another item fails, material already debited still opens its liability.

Conservation by good (PE3). For every actual history h and good g,

$$\sum_{a:\, g(a)=g} b_a(h) + \mathrm{Paid}_g(h) = \mathrm{Damage}_g(h), \qquad \mathrm{Paid}_g(h) \leq \mathrm{Signed}_g(h).$$

Paid quantities count only eligible receipts consumed by the allocator at actual material positions. Induction appends one input: first post its new losses, then allocate eligible delivery to original rows. The local identity “remaining after allocation plus the new allocation equals remaining before

allocation" adds to the preceding conservation identity. Earlier eligible rows have priority, and total credit assigned at one delivery position cannot exceed the signed quantity at that position.

Double unilateral deadlock, non-reduction bound, and widest safe guard (PE3, PE6). Under the old single-signature receipt rule without an independent victim-acceptance channel at the same position, a payer's unilateral order credits no cross-community debt (on the six-unit source, payer-signed `returnAt 6` moves stocks to $(117, 107)$ with zero credit while $[(\rho_0, 6)]$ remains intact, `after_the_payer_returns_six_the_original_six_is_still_in_the_whole_account`), while every positive cross-community unilateral receipt clears old debt only by simultaneously opening an equal reverse cross-community loss at the same material position *pos* (victim-signed `receipt 6` closes ρ_0 to 0 and writes `(reverseOccurrence, 6)` at that position, `the_victims_receipt_really_closes_the_old_six_and_writes_the_reverse_six`; generally `every_cross_community_receipt_opens_its_equal_reverse_damage`). Because one material item cannot spend the same good twice, total credit assigned to good g at *pos* is bounded by that receipt's quantity and hence by the new good-g loss filed at *pos*; summing across distinct positions of one input gives $\text{Assigned}_g \le \text{Filed}_g$. Combining this with step conservation shows that for every prefix h and every continuation script u,

$$\text{Out}_g(h) \le \text{Out}_g(h +\!\!+ u).$$

On the original six-unit history h_0, $6 \le \text{Out}_{\text{false}}(h_0 +\!\!+ u)$ holds throughout, while retaining the old production law to clock 5 yields zero food delivered against unit demand and leaves no independent exit under any control map, establishing dual material and receipt obstructions. Once the cross-community damage guard is installed, it forwards a request if and only if it is originally lawful with zero new cross-community damage, containing every safe admission predicate as a subset.

A **continuation step** consists of a registered common word actually paid from the current inventory, together with an independently payable exit word for each listed participant. Every common-word item respects both original and current subsistence and maintenance floors, and the actual duration is positive. Finite paths concatenate these steps. For a set R of states, define

$$\Phi(R) = \{x : \ \forall B \in \mathbb{Q},\ \exists \text{ a finite actual path } x \rightsquigarrow y,\ y \in R,\ t_y > B\}.$$

The largest viable region is

$$V = \bigcup \{R : R \subseteq \Phi(R)\}.$$

Its defining objects are finite paid paths that return to the region beyond every original-clock bound.

Viability theorem (PE6).

$$x \in V \quad \Longleftrightarrow \quad \text{a registered, paid, clock-cofinal trajectory starts at } x.$$

For necessity, take the set of all states of the trajectory. For sufficiency, select a shortest actual path beyond the current target clock and consume it step by step. At its endpoint, reset the target and select again from that same successor. Induction on integer clock increments reaches every rational bound after every prescribed cut. Actual successor identities concatenate the paths, carrying subsistence and independent exit with them.

Fix an original loss occurrence, independent receipt channels, registered submissions, and approved production alternatives. Let x_h be the actual state reached by a prefix h. Define

$$\begin{aligned}\mathrm{Entry}(h) \iff\ & h \text{ passes original admission and its pending work and review entry are ready, and} \\ & \exists c, m:\ c \text{ is an approved production choice,} \\ & m \text{ is paid from } x_h \text{ and meets the first-trial requirement of } c, \quad m(x_h) \in V_c.\end{aligned}$$

The optional production choice c may be empty, retaining the current rule and original roster. Choosing an alternative requires original approval, and m executes that alternative's literal first trial. The region V_c is determined by the chosen execution law, original subsistence baseline, and roster. The actual first-trial successor belongs to V_c, and continuation begins there.

The viable whole-ledger settlement domain is

$$\mathcal{S} = \{h : \mathrm{Entry}(h) \ \wedge\ \forall a \in L(h),\ b_a(h) = 0\}.$$

Whole-ledger terminal theorem (PE6–PE8). The construction returns an actual whole-ledger settlement exactly when $\mathcal{S} \neq \emptyset$. Its selected $h_* \in \mathcal{S}$ satisfies

$$|h_*| = \min\{|h| : h \in \mathcal{S}\}.$$

The whole-settlement report and the whole-portfolio optimum report select the same literal prefix and the same actual realization. If $\mathcal{S} = \emptyset$, every actually viable candidate retains a positive original row. If no viable entry exists, each candidate retains its original admission or viability obstruction.

The proof minimizes the whole-portfolio indicator $\mathbf{1}_{\{\exists a,\ b_a(h)>0\}}$. This tests zero balance row by row without adding goods with different units. A clearing prefix attains zero; minimization of natural-number length then selects the first attainable visit count. Conversely, an actual zero-balance output is itself an element of $\mathcal{S}$. Optimization for one original claim separately uses its lower-bounded real value set and distinguishes an attained infimum, strict approximation at every positive tolerance, and an empty domain.

Visit count and physical time have separate frontiers. Over the viable settlement domain $\mathcal{S}$ equipped with joint production trials, put

$$T_* = \inf\{t_{x_h} : h \in \mathcal{S}\}.$$

When this domain is nonempty, every $T > T_*$ admits an actual settlement prefix completed before T. If T_* is attained, select the fewest visits among prefixes at that time. Otherwise every strictly later deadline has an actual-prefix witness.

Compulsory convergence, unique peace tail, and anti-filibuster hourly progress (PE6). Under the constitutional terminal construction on the six-unit source, regardless of competing interleavings, every lawful continuation converges at visit 4 (physical clock 5) to the same peaceful whole-ledger resident with a unique continuation tail ($\exists!$ tail, `source_generated_termina l_peace`), preserving separate financial principal 6 alongside cleared $[(\rho_0, 0)]$. Along continuation, integer hours are never skipped, no competing interleaving can delay due subsistence past the

current priority queue (crossing into the next hour within pending$(x_n)+1$ visits, `the_due_service_cannot_be_delayed_past_the_pending_queue`), and the first visit crossing each open hour $h+1$ pays at least 1 unit of food to every member (`the_first_crossing_of_each_open_hour_is_paid`).

B. Reattachment thresholds, observation-fibre duality, production bounds, and payment at the original position

B.1 Strict reversal, reattachment obstructions, and observation-fibre duality

An executable menu M consists of candidates with a score u, and has frontier

$$v(M) = \sup\{u(w) : w \in M,\ w \text{ is actually executable}\}.$$

Suppose the original war and peace menus and the reattached war and peace menus are each nonempty and bounded above. Strict reversal

$$v(W_0) > v(P_0), \qquad v(W_1) < v(P_1)$$

is equivalent to two actual witnesses with uniform positive gaps:

$$\exists w \in W_0, \delta > 0,\ \forall p \in P_0,\ u(p) + \delta \le u(w),$$

$$\exists p \in P_1, \varepsilon > 0,\ \forall w \in W_1,\ u(w) + \varepsilon \le u(p).$$

The forward direction selects an actual candidate inside the strict gap between frontiers. The reverse direction bounds the opposing frontier using the uniform gap. The suprema need not be attained. Empty, unbounded, attained, and strictly approached menus have their respective dispositions. (PE1)

Under incomplete observation $\omega : \mathcal{W} \to \mathcal{O}$ ($\mathcal{W}$ nonempty), write $\mathcal{F}(s) = \{b \in \mathbb{R} : \exists a,\ \forall w \in \omega^{-1}(s),\ a \text{ is executable and } u(w,a) \ge b\}$ for local fibre floors and $\mathcal{P} = \bigcap_s \mathcal{F}(s)$ for global prior-policy floors. When $\mathcal{P} \neq \emptyset$ and every $\mathcal{F}(s)$ is bounded above, the robust frontier satisfies

$$\sup \mathcal{P} = \inf_{s \in \mathrm{range}(\omega)} \sup \mathcal{F}(s), \qquad b_* \in \mathcal{P} \iff \forall s,\ (\sup \mathcal{F}(s) = b_* \Rightarrow b_* \in \mathcal{F}(s)),$$

where $b_* = \sup \mathcal{P}$. Inclusion $\mathcal{P} \subseteq \mathcal{F}(s)$ gives $\le$; any strict gap c would lie below $\sup \mathcal{F}(s)$ on every fibre and hence inside every $\mathcal{F}(s)$ by downward closure, placing $c \in \mathcal{P}$, a contradiction. Non-binding fibres ($b_* < \sup \mathcal{F}(s)$) likewise contain b_* by downward closure, isolating attainment on binding fibres. (PE1)

For the same payable conflict word and peaceful benchmark, reattachment at rate λ yields the original margin $m - \lambda T$ ($T \ge 0$). The native three-party material control exposes private gain under negative-sum war and coalition stock invariance (`exact_native_material_controls`, `a_paid_return_has_exact_coalition_incidence`): peaceful stocks $(4, 7, 10)$ total 21, shifted war $(8, 7, 4)$ drops total social stock to 19 (net destruction -2) while giving the initiator $+4 = -2 + 6$ by externalising 6 units, and full reattachment $(2, 7, 10)$ conserves total post-war stock across the two coalitions at 19 while returning the -2 loss to the initiator. Full reattachment strictly reverses the

comparison exactly when $0 < m < T$. For a finite family, an eliminating rate $\lambda \in [0,1]$ exists if and only if $\lambda = 1$ eliminates all positive margins; infeasibility holds if and only if some row has unshifted positive gain ($T_i = 0, m_i > 0$, blocking every rate λ) or insufficient shift ($0 < T_i < m_i$). When feasible, define

$$\lambda_* = \max(\{0\} \cup \{m_i/T_i : T_i > 0\}).$$

For nonnegative rates, all margins are nonpositive exactly when $\lambda \geq \lambda_*$; every smaller rate leaves a positive-margin candidate, and $\lambda_* > 0$ produces a binding tie on a positive-shift row. Strict negativity throughout holds if and only if every zero-transfer row already has $m_i < 0$ and every positive-transfer row has $m_i/T_i < \lambda$ (equivalently $\lambda > \lambda_*$ when $\lambda_* > 0$, whereas when $\lambda_* = 0$ strict negativity already holds at $\lambda = 0$; `strict_exclusion_exact`, `strict_after_positive_minimum`). When payment capacity caps payable rates at $\lambda_{\text{cap}} = \min_j c_j \in [0,1]$, payable war-eliminating rates form the exact closed interval $\Lambda_{\text{peace}} = [\lambda_*, \lambda_{\text{cap}}]$ (`joint_interval`); whenever $\lambda_{\text{cap}} < \lambda_*$, every $\lambda > \lambda_{\text{cap}}$ incurs a named material shortfall while every $\lambda \leq \lambda_{\text{cap}}$ leaves a positive war margin (`source_minimum_or_named_shortage`, `strict_source_rate_exists_iff`), establishing the judgment-proof obstruction where ex post liability alone cannot deter war. When authority or capacity changes with amount, the literal word is checked again at each rate and intersected with $\lambda \geq \lambda_*$. (PE2)

B.2 The exact production-period bound

Let $c = (p, g)$ identify a payer and good. Write D_c for all original debt drawn on that cell, $s_c \geq 0$ for spendable stock after reserving production needs, and $r_c \geq 0$ for the actual net gain per period. For a finite original ledger,

$$\exists n \in \mathbb{N}\ \forall c, \quad D_c \leq s_c + n r_c \quad \iff \quad \forall c,\ s_c < D_c \Rightarrow r_c > 0.$$

Under the condition on the right,

$$n_* = \max_c \left\lceil \frac{D_c - s_c}{r_c} \right\rceil_+$$

satisfies $n \geq n_*$ exactly when stock at period n covers every debt. Cells already covered contribute zero. The proof applies the ceiling identity at each cell and takes the maximum over the finite ledger support. With original funding release and independent receipt channels, the construction selects the earliest actually funded cut, pays by original key, and restarts the same production from its retained inventory. (PE7)

B.3 Both parties sign the same actual debit

Let $n_0(p,g), n_y(p,g)$ be subsistence needs at the original cut and actual successor y, and let $m_0(p,g), m_y(p,g)$ be their maintenance needs, all read by the operative law. The dynamic protected reserve is

$$R_y(p,g) = \max\{0, n_0(p,g), n_y(p,g)\} + \max\{0, m_0(p,g), m_y(p,g)\}.$$

The condition $R_y(p,g) \leq s_y(p,g)$ holds exactly when the stock can be divided into nonnegative subsistence and maintenance shares, each covering both original and current needs. Compare the

two maxima componentwise for the proof. The two services consume separate shares, and every actual intermediate inventory is checked this way. (PE16)

At an actual state x and original debt key $k = (v, p, g)$, a positive rational payment q is funded when

$$\begin{gathered} v \neq p, \quad 0 < q \leq b_k(x), \quad \mathrm{Valid}_x(\mathrm{pay}(k, q)), \\ \mathrm{Answer}_p(x, \mathrm{ord}(x), \mathrm{pay}(k, q)) = \mathrm{true}, \\ \mathrm{Answer}_v(x, \mathrm{ord}(x), \mathrm{pay}(k, q)) = \mathrm{true}, \\ \forall z, g', \quad s_{\mathrm{pay}(x)}(z, g') \geq \mathrm{Required}_x(\mathrm{pay}(x))(z, g'). \end{gathered}$$

Both answers read the same ledger resident, actual ordinal, current stock, and literal transfer. The dynamic reserve is evaluated at the actual post-payment state. The executable checking entry takes a finite enumeration covering every address it checks; the mathematical payment construction can also consume the fundedness proof directly.

On success the allocator consumes the position just written and produces

$$b_{k'}(x') = b_{k'}(x) - \mathbf{1}_{\{k'=k\}} q.$$

All original addresses remain. A production revision first obtains the affected original parties' approval and executes its trial; the trial's output then enters the actual inventory checked for payment. Refusal, unauthorized action, overpayment, and material shortage each generate an original-ledger successor with a specific reason. (PE16–PE17)

Let $\mathcal{A}_k(x) \subseteq \mathbb{R}$ be the embedded set of all funded rational amounts and let $q_* = \sup \mathcal{A}_k(x)$. When nonempty, $q_* \leq b_k(x)$ is its exact upper bound. Attainment generates the actual payment. Nonattainment generates, for every $\epsilon > 0$,

$$q_* - \epsilon < q < q_*,$$

together with the actual successor for q. The empty case supplies a named refusal for every amount. If the full original balance is funded, $q_* = b_k(x)$ and the construction selects it. (PE18)

A receipt cannot be credited twice. The first component of each actual receipt address is the ordinal at its creation. At step n of any dispatch history, the ordinal is $\mathrm{ord}(x_0) + n$. Reusing one receipt at $i < j$ would make its address equal to two different natural numbers. Production revisions and refusals also consume their own ordinals, so the argument covers every interleaving of the three submission classes. (PE17)

C. The actual settled state and the open language

C.1 Opening peace from actual inventory

At the settled state x_*, check a common production word w and an independent exit word e_p for every listed participant. The checks include registration, itemwise authorization, original and current subsistence and maintenance floors, nonnegative accumulation with at least one positive net gain, and positive actual duration $q > 0$. Repetition uses certified stability: shifting the clock does

not change action effects, adding nonnegative reserves preserves operative rules and registration, and duration is preserved under these transformations.

Successful checks generate the running entry from the entire original ledger, actual stocks, and operative production rule at x_*; initial language coverage is empty. Taking x_* as the common reserve base, for every public-history-driven strategy π and every n, the next transition actually produces, pays subsistence, preserves each independent exit and every settled original row, emits zero new loss at every actual item, and strictly increases at least one stock coordinate. For every starting cut a and elapsed number of rounds n,

$$t_{a+n} \geq t_a + nq.$$

This gives clock cofinality after every cut. (PE13, PE19)

The original six-unit consumer has debt-material inventories, written as "victim/payer,"

$$(111, 113) \xrightarrow{\text{approved new-rule trial}} (115, 117) \xrightarrow{\text{six-unit delivery signed by both}} (121, 111).$$

The trial and settlement states both have clock 5. The payer's actual 111 directly enters the continuation and exit checks. The original address

$$a_0 = (2, 0, (\text{false}, \text{true}, \text{false}))$$

remains with zero balance. The first paid proposal starts at clock 5, exposing a protected-term change and reaching clock 6. The second starts at clock 6, exposing removal of an original personal exit and reaching clock 7. Each pays half a unit of food. A third-meal counterexample identifies the payer's food shortage $0 < 1$ at actual position 2, retaining the same settled ledger. (PE19)

The earlier PE14 consumer uses a different settled entry: personal exit is exposed first and public terms second. Both visits likewise debit half a unit and retain the original debt address. Each history retains its own order.

C.2 Expressive failure, least extension, and anti-fragile invariants

The effect type includes "an originally payable named continuation ceases to be payable under the proposed rule" and "a protected public term changes." Effects are determined at the actual pre-state of the paid visit and its literal proposed rule. Two candidate institutions can share the exact same zero-debit old offer word `oldOfferWord` or identical old tariff quote while one preserves and the other strips an originally payable personal exit, so no classifier reading only the old word or old quote can detect stealth capture ($\neg\exists\,\text{classify},\ \forall q,\ \text{classify}(\text{obs}_{\text{old}}(q)) = \text{losses}(P(q))$, `no_exact_loss_classifier`, `old_offer_word_cannot_classify_loss`). (PE11)

Represent the refusal language by (Code_n, D_n) with coverage $C_n = \bigcup_{c \in \text{Code}_n} D_n(c)$, and let E_n be the effects witnessed at round n. A language revision is generated exactly when

$$E_n \setminus C_n \neq \emptyset.$$

Revision forms $\mathrm{Code}_{n+1} = \mathrm{Code}_n \oplus \mathbf{1}$, embeds every old code with unchanged meaning $D_{n+1}(\mathrm{inl}\, c) = D_n(c)$, and sets $D_{n+1}(\mathrm{inr}()) = E_n \setminus C_n$. Hence

$$C_{n+1} = C_n \cup E_n, \qquad C_{n+1} \subseteq B \iff C_n \subseteq B \ \wedge \ E_n \subseteq B.$$

This is exact leastness of coverage. Induction from empty initial coverage gives

$$e \in C_n \quad \iff \quad \exists i < n,\ e \in E_i.$$

Every new expression has an original paid witness, and each earlier exposure remains expressible. Over the full effect space $\mathcal{E}$, the unsealed vulnerability set $U_n = \mathcal{E} \setminus C_n$ shrinks strictly ($U_{n+1} \subsetneq U_n$) at every novel exposure (likewise shrinking strictly on the `Party` $\times$ `Good` sub-key domain of adaptive peace, bounding novel-exposure count whenever that key domain is finite), while preserving the entire native execution surface—so public compilation of every finite native word `word` and every participant–good productive gain remain identical. (PE10, PE12)

The ledger resident generated by this language write is exactly the resident used by the next material round. The same write carries original rows, rights to request, and budgets; the new direction enters with zero budget. The next production reads the original tariff and orders at this state. Least revision thus has both a semantic conclusion and an actual subsequent consumption. (PE13)

At every round the proposal syntax may depend on the round, public history, original clock, public terms, and current language. An interpreter maps that syntax into the same execution input; failed interpretation takes the original due continuation. Over this universally quantified strategy domain, the invariants and the equivalence for revision hold at every round. (PE11–PE13, PE19)

C.3 The current source of an archived obstruction

Revision retains obstructions from all old supports in its archive. Let S and S' be the old and new support sets, with restriction map $b : S' \to S$. An obstruction o at old support s has archive address (s, o). For an active new support a,

$$(s, o) \text{ is a current local obstruction} \quad \iff \quad s = b(a).$$

Necessity reads the first component of the archive address. For sufficiency, equality of the supports lets the same original obstruction serve locally. The complete archive persists, while the next actual disposition is generated by an obstruction at the active support's restriction. (PE13)

D. Review costs, zero-fee erasure, privacy, and delayed evidence

Fix filing work and per-request review work at an actual public cut. Each work item is checked for authorization, debit bearer, and material flow. Sum the actual per-good debits of these admissible procedures to obtain the charge vector c. For any fully covering charge vector b,

$$b_g \geq \mathrm{cost}_g \quad (\forall g) \quad \Longrightarrow \quad c_g \leq b_g \quad (\forall g), \qquad c_g = \mathrm{cost}_g\,.$$

Leastness is relative to these checked work procedures. The original dispatcher preserves the common production slot, so this tariff's production-slot opportunity component is zero. A positive filing amount, valid payment order, and common budget admit a single debit at the next actual material position; insufficient funds or a charge exhausting the productive gain takes the original production continuation. (PE15)

Against ordinary continuation at the same clock, each paid review has strictly negative material payoff for its actual payer under every strictly positive per-good weighting. Splitting a submission into paid fragments counts each original ordinal once and repeats the filing charge for each fragment; routing an already covered scope through a distinct proxy payer (`delegated.payer` $\neq$ `levy.signer`) likewise debits the half-unit food fee from the actual payer and returns the fixed refusal without opening a second review (`a_different_filer_cannot_reopen_the_same_scope`). Presentations producing no actual charge can be erased from the entire infinite state–observation trace without changing any action-facing information fibre; any changed readout traces back to an earlier actual material write authorised by its payer. (PE8–PE10)

Two-history prepayment privacy (PE9, PE21). Compare two actual openings with the same reserve base, tariff, and payment/proposal orders. Hidden inventories above that base and private qualification answers may differ. Fix the same public strategy and require the same public due agenda whenever observations agree. Every prefix before an actual charge has the same public transcript and language. The first new distinction requires the same bill to have been actually debited on both sides.

Induction on the prefix proves this result. Common observation and agenda select the same input and payment branch. Before payment, both replies equal the original public agenda and both languages persist. If the next prefixes differ, the unpaid branch is impossible, so both visits have receipts for the same literal bill. The theorem locates the first difference at actual payment while retaining each side's actual successor.

Across rebases, a shared reserve relation retains the original stock lower bound, financial claims, and technology coordinates. The refresh acknowledgement factors through the pre-write agenda, whose announced landing clock equals the actual landing clock. If the new published supply terms differ, the actual production words executed on the two sides differ; new supply terms are released with actual production. (PE25)

Delayed evidence on the original payment (PE20, PE25). When charging has occurred but coverage evidence remains incomplete, retain the original receipt, exact effect scope, and actual material successor. Completed evidence reads that scope at the state actually reached. If it is already covered, the successor is that same state and there is no new language write. Otherwise the least extension is written there. Read paths across actual due payments and rebases preserve the same original scope; completion consumes the original receipt. Completion uses the existing payment without charging the original review fee again; its original orders remain evidence of that paid occurrence.

A finite adaptive duration observation cannot determine all protected terms or decide the paid scope: for any adaptive reader querying durations on a finite step set S_Q, set the diagonal clock $t^\dagger = \max(\{0\} \cup \{\tau(s) : s \in S_Q\}) + 1$ and keep the original readings on S_Q while altering a protected term at $t^\dagger$, yielding an institution with identical S_Q-observations and a nonempty paid scope (`e`

`very_finite_duration_reader_has_a_hidden_protected_change`, `no_finite_duration_reader_decides_the_paid_scope`). This diagonal construction identifies the exact object for which evidence is still needed. (PE20)

E. Continuation before settlement, generated production words, and the nine-stage deployed track

At an unsettled state with a checked continuation recipe, each round first executes current subsistence and independent exits, then reads a payment or production-revision request. For every adaptive strategy and finite waiting period, original addresses persist, refusal retains debt, and physical time satisfies $t_n \geq t_0 + nq$. When both signatures arrive later, payment is checked against that round's actual inventory after the due step. Sufficient stock pays the original debt and the settled state enters the network directly. (PE23)

Production repair checks registered recipes first, followed by one round of named edits. Given a finite original registered alphabet and a finite participant roster, after both layers are exhausted it jointly searches for a common production word w and each person's exit word e_p. Define width by

$$\mathrm{width}(w, (e_p)) = \max(|w|, \max_{p \in P} |e_p|).$$

At a fixed current state every finite width is completely checked. A fresh feasible joint recipe exists exactly when search succeeds at some finite width. The first successful width carries named obstructions at every narrower width. For the proof, place any feasible recipe in its finite width layer; rejection of that layer directly contradicts its production or personal-exit conditions. (PE24)

The running entry also checks actual duration, its registered basis, and the current search window. A feasible word within the window forces the first successful-width actual production write. Exhausting the window still pays the original due step and retains the checked layers' obstructions. Actual ordinals make the available width unbounded. In the original six-unit deployment, an empty-word failure at clock 7 leads to the generated width-four write at clock 8, increasing each person's debt material by two. From the same post-trial state (`source`, clock 5, debt 6), `Deployment.{Facade,Ingress}` advances deterministically through `due` (clock 5 → 6, debt 6), `refusal` (clock 6, debt 6), `settlement` and `installation` (clock 6, debt 0), `pendingReview` (clock 7), and `rebase` (clock 8): from there, the unwaited batch schedule (`Facade.Stage`) immediately executes `revision` at clock 8 (admitting evidence only once) and five `renewal` rounds to clock 13 (matching all 13 local observations between batch and stepwise processes), whereas the persistent online entry (`Ingress`) runs two waiting due steps across clocks 9–10 (`waiting 1, 2`), cold-recovers and writes U8 at clock 10 (`revised 2 0`), and completes two renewals across clocks 11–12 (`revised 2 2`, cursor 11) to produce the exact clock-12 parent head (`013bd261...`) sealed for four-host admission. Both schedules and the immediate-revision branch in C.1 (clocks 5 → 6 → 7) retain separate literal identities from the same trial state. (PE24–PE26)

F. External write-by-write factorization and historical identity

Let the original history consist of states x_n and actual writes w_n. An external presentation supplies states y_k and optional writes e_k. Set

$$o(0) = 0, \qquad o(k+1) = o(k) + \mathbf{1}_{\{e_k \neq \emptyset\}}.$$

The local check requires an empty write to preserve $y_{k+1} = y_k$. A nonempty write must be exactly $w_{o(k)}$ and land at $x_{o(k)+1}$. A common origin and all local checks hold exactly when there is a unique factorization f preserving states, writes, and ordinal recursion. Necessarily $f = o$: induction from zero and increments of zero or one determines the index, and local-check induction determines the states. (PE8)

External progress

$$\forall n \; \exists k, \quad n \leq o(k)$$

is equivalent to a unique first visit to every original cut. Without progress there is a last cut and all later writes are empty. Failed checks locate a foreign origin or the first mismatching write, retaining the already matching prefix.

Two direct counterexamples exhibit different failures. The first renews aggression after the common opening, then continually pays subsistence, retains settlement of the old rows, and has a cofinal clock; the renewed aggressive write differs from the original. The second merely reverses the order of two harmless supply sequences:

$$(\text{true}, \text{true}, \text{false}, \text{false}) \quad \text{and} \quad (\text{false}, \text{false}, \text{true}, \text{true}).$$

The complete raw material states and clocks agree at every visit; independent exits remain, no new loss is opened, and the old ledger is cleared. Their actual-item orders differ, so they cannot factor through the same fixed order. Write identity retains information about actual occurrence that endpoint readings erase. (PE8)

G. Ending war, unprioritized stall countermodel, independent renewal, and irreversible deadlines

For a given observation function, states with the same visible history form an observation fibre. A sustained protective strategy exists exactly when a protective family contains the opening and each reachable fibre has one shared civilian action whose every original opponent response remains in that family. The opponent may observe the civilian action before responding. Every opponent prefix admits an independent exit under the same strategy. At an intermediate position of a submitted civilian word, the checked suffix remains executable before a fresh exit is paid. Necessity takes the positions reachable under a successful strategy. Sufficiency chooses the shared action at each fibre and inducts on rounds. (PE4)

Each failed strategy has a finite first failure carrying the original whole ledger. With clock cofinality required, failure is exactly a finite first counterexample or a trajectory that passes every local visit while its original clock remains bounded. In the signed constitutional game, no civilian

strategy forces ceasefire and withdrawal completion if and only if every signed policy suffers an allowed-play dead end, a finite joint breach, or indefinite unpaid delay; dual to $\exists\pi\,\forall\alpha$, whenever one uniform opponent policy defeats every signed civilian policy ($\exists\alpha\,\forall\pi$), tick $t_* + 1$ immediately following the first attack automatically files a live original reparation row in the whole ledger (`the_uniform_attacker_files_the_first_original_loss_against_every_signed_policy`). (PE4–PE5)

Under the installed emergency law, admitted withdrawal, a closed protective observation family, and nonempty lawful local menus, choose one strategy before quantifying over opponents and lawful interleavings. Under priority scheduling, a checked literal commitment with N original pending entries and body length ℓ completes at visit $N + \ell$ with absolute endpoint clock independent of competing interleavings. Conversely, without priority scheduling, when pending work exists and review is available, lodging withdrawal at tick 0 followed by `stopReview` from the opponent at every tick ≥ 1 emits an empty material inventory forever and keeps the clock at $\tau_0 < \tau_0 + 1$, leaving withdrawal incomplete at every tick. After revocation, subsequent authority exists if and only if an actual independent renewal record exists; no single signer can regenerate it alone. A continuously refusing exposed bearer retains their stocks, reparation rows, and released mandate against any adaptive coalition, and any subsequent paid controlled action traces back to an earlier paid `.propose` item by a task member and each original bearer's affirmative `true` answer at that material grant item. (PE5)

Emergency authorization checks every rational candidate expiry against actual issuance, service, both independent review payments, and each personal exit, producing the feasible domain and its earliest and latest frontiers. The PE22 consumer issues at clock 1, expires at 3, and serves through 4 with two unit review payments; missing the second payment or a personal exit produces distinct named refusals. Once $T_{\mathrm{dead}} < t_x$, because no lawful original script runs the physical clock backwards, every subsequent input suffix preserves $T_{\mathrm{dead}} < t_y$ and permanently refuses every `.propose`, `.accept`, `.renew`, and protected `TaskUse` item without possibility of reset. (PE5, PE22)

H. Fixed versions and proposition correspondence

The source root is `Lean/SaturationMonoid/ProcessGame/Society/Institutions/Conflict/`, abbreviated **C**. Further abbreviations are `T=C/Constitution/TerminalPeace`, `U=T/Universal`, `F=U/Terminal/FixedPoint`, `O=F/OpenPeace`, `J=U/Terminal/Productive/Clearance/Joint`, `X=J/Reinterpreted`, `R=O/Network/Runtime`, and `E=C/Emergency/Installation`. Module and declaration names below are exact source identifiers.

Fixed group	Homework version	Original acceptance identity
PE1–PE10	`7e4da746f300b3796b8b18afb26cb0765c29afcb`	Separate original receipts for reparations, sustained stopping, cofinality, joint settlement, prepayment, and adaptive revision
PE11–PE14	`b675168bda8f52a6565a8db3617e262daa738bb0`	Open language `4dc2e25393`, network first write `369df502de`, recursion `4caa9fd0a2`
PE15–PE22	`71cb159f4f0727d8fb67e7a436a1476e03a0d913`	Actual work `b867cdf377`, reinterpreted settlement `5c649ff440`, actual reserve base `71cb159f4f`
PE23–PE26	`3a90cba5ac7427737ee2fe630612dfed43e45da1`	Interleaved due `46afb0dec1`, deployment `f3014e5e63`, generated production `c58d788478`, joint refresh `f4578baa82`

Fixed group	Homework version	Original acceptance identity
Intake ceiling	`cf880bb282f0ff28d6e9f9fc890341baaa8b0eb0`	PE27–PE33 execution, recovery, and disposition are mapped in the execution appendix

H.1 Producing declarations, direct consumers, and decisive comparisons

PE	Producing declaration or proof	Direct consumer and use
1	`C/Choice/Frontier.lean`: `strict_flip_iff`; `C/Choice/Robust.lean`: `robust_frontier_eq_worst_observation`, `robust_frontier_attained_iff_binding_fibers_attain`; `C/Root/{Menus,History,Cofinal,Policy,Restitution}.lean`: `a_paid_return_has_exact_coalition_incidence`	`C/Consumers/{Menu,Timing,Native}.lean`: `exact_native_material_controls`, $+4, 0, -2$ at the same cut, totals 21/19/19, and observation-fibre guarantees
2	`C/Threshold/Reattachment.lean`: `exact_upper_ray`, `no_admissible_rate_exact`, `any_positive_unshifted_blocks_every_rate`, `strict_exclusion_exact`; `C/Threshold/Capacity.lean`: `joint_interval`, `source_minimum_or_named_shortage`, `strict_source_rate_exists_iff`; `C/Root/RateDomain.lean`	`C/Consumers/{Native,Exact,Endpoint}.lean`: original slot, attained 2/3, capacity interval, judgment-proof obstruction, and overstock/repeated-settlement refusal
3	`C/Reparation/Actual/Root/Batch.lean`: `every_affected_address_is_filed_on_the_same_generated_next`; `Run/Conservation.lean`: `the_original_whole_ledger_conserves_each_material_good`; `T/Material/{Payer,ReceiptAccount,Outstanding}.lean`: `after_the_payer_returns_six_the_original_six_is_still_in_the_whole_account`, `the_victims_receipt_really_closes_the_old_six_and_writes_the_reverse_six`, `every_original_history_keeps_or_increases_outstanding_reparation`; `U/Protection/Guard.lean`: `the_guard_forwards_exactly_the_safe_original_requests`	`Consumers/{Joint,Registration}.lean`; `C/Reparation/Consumers/ActualDamage.lean`; double unilateral deadlock and widest safe guard `any_safe_original_admission_is_contained_in_the_guard`
4	`C/Constitution/Peace/Contract.lean`: `sustainable_civilian_peace_iff_one_move_protects_each_reachable_observation_fibre`; `a_cofinal_policy_fails_iff_first_counterexample_or_an_original_clock_bound`	Root, Failure, and Shortage in the same directory: actual opponent prefixes, first failure, and independent exit
5	`C/Constitution/ForcePeace/Contract/{Root,Delay,Defeat,Adversary}.lean`: `peace_is_enforceable_and_war_authority_cannot_be_unilaterally_reborn`, `a_lodged_withdrawal_has_an_actual_infinite_delay_without_priority`, `no_civilian_strategy_forces_peace_iff_each_signed_policy_has_an_original_obstruction`, `the_uniform_attacker_files_the_first_original_loss_against_every_signed_policy`; `ExposureGuard/Contract/Root.lean`; `C/Civil/Mandate/Root.lean`; `Commitment/Timing.lean`: `initial_native_rank_and_body_length_give_the_sharp_bound`	`ExposureGuard/Commitment/Consumers/Installed.lean`; `ExposureGuard/Consumers/Renewable/Dispatcher.lean`: coalition refusal invariants, uniform-adversary auto-filing, and revival proposal trace

PE	Producing declaration or proof	Direct consumer and use
6	`U/Runtime/Viability/Strategy.lean`: `live_iff_cofinal_trajectory`; `U/Terminal/Whole.lean`: `simultaneous_clearance_is_generated_exactly_when_a_live_original_prefix_pays_every_row`; `T/Root.lean`: `the_original_source_has_independent_material_and_receipt_obstructions`; `T/Renewal/Cofinality.lean`: `the_due_service_cannot_be_delayed_past_the_pending_queue`, `the_first_crossing_of_each_open_hour_is_paid`	`T/Completion/Root.lean`: `source_generated_terminal_peace`; visit-4 compulsory convergence, unique peace tail, anti-filibuster hourly step, and unilateral ≥ 6 bound
7	`U/Terminal/Productive/Clearance/Budget.lean`: `fundingCycles_le_iff_funds`; `Compiler.lean`: `finite_original_losses_are_cleared_by_production`; `J/Compiler.lean`: `compileJoint_pays_iff`	`Clearance/Consumers/OriginalBatch.lean`; `J/Consumers/Settlement.lean`: reversed exchange is short by six
8	`F/{Compiler,Closure}.lean`; `U/External/Trace.lean`: `exact_local_checks_iff_unique_factor`; `U/External/Frontier.lean`: `progress_iff_every_original_cut_has_unique_first_visit`	`F/NoFreeProbe/Prepayment/Consumers/JointOpening.lean`; `U/External/Countermodel/{Cofinal,SafeOrder}.lean`
9	`F/NoFreeProbe/Influence.lean`: `zero_tenders_leave_every_information_fibre_unchanged`, `any_changed_read_traces_to_a_payers_actual_write`; `RepeatedCharges.lean`; `Prepayment/Run.lean`: `locate_the_first_paid_distinction`	`Prepayment/Consumers/{Original,JointOpening}.lean`: half-unit prepayment; a charge exceeding net gain takes the same renewal
10	`F/AdaptivePeace/Contract.lean`: `every_novel_attack_generates_stronger_peace`, `Step.strictly_shrinks_unsealed`, `Step.keeps_every_native_word`; `History.lean`: `coverage_has_exactly_its_original_paid_witnesses`, `an_old_scope_cannot_open_another_review`	`AdaptivePeace/Consumers/{Original,JointOpening}.lean`: `a_different_filer_cannot_reopen_the_same_scope`, original levy repair, paid resubmission, and proxy-filer defence
11	`O/Boundary.lean`: `paidLosses_eq`, `no_exact_loss_classifier`, `old_offer_word_cannot_classify_loss`; `O/{Ingress,Dispatch,Payment}.lean`	`O/Consumers/Original.lean`: `a_new_policy_removes_an_exact_paid_exit_from_the_same_word`, `the_old_quote_does_not_classify_the_new_continuation_loss`
12	`O/Language.lean`: `least_repair_iff`, `denotes_inherited`, `denotes_fresh`; `O/Repair.lean`; `O/Universal.lean`: `open_future_permanent_peace`	`O/Consumers/JointOpening.lean`; `O/Network/Compile.lean`: the original paid witness for every expression
13	`R/Advance.lean`: `next_reads_generated_first_write`; `R/History.lean`: `the_revised_resident_drives_the_next_material_write`, `history_is_living_and_peaceful`, `history_clock_bound`; `R/LocalU7.lean`: `archived_obstruction_is_local_iff`	`R/Consumers/Original.lean`: the revised resident drives the second material write
14	`O/Network/Consumers/Original.lean`; `R/Consumers/Original.lean`: `two_paid_gaps_reenter_the_material_dispatcher`	Two half-unit debits; covered resubmission, wrong signer, and stale-order comparisons
15	`F/NoFreeProbe/Pricing/Quote.lean`: `the_generated_fee_is_the_least_full_charge`; `Pricing/Entrance.lean`: `installation_reads_the_same_cost_at_every_epoch`	`Pricing/Consumers/{Work,Original}.lean`: checked work, insufficient payment, and original production
16	`J/Revision/{Reserve,Technology,Protected,Portfolio}.lean`; `O/Operational/Certificate.lean`: `verifyPrepared_accepts_every_admissible_word`	`O/Operational/Consumers/{Certificate,OriginalInventory}.lean`: full paid word and alternative ordering

PE	Producing declaration or proof	Direct consumer and use
17	`X/Funding.lean`: `checkFunding_exact`; `X/Payment.lean`: `PaidWrite.whole_successor`; `X/History.lean`: `every_history_preserves_its_original_addresses`, `no_receipt_can_be_booked_again_in_the_same_history`	`X/Consumers/{OriginalTrial,OriginalPayment,OriginalSource}.lean`
18	`X/Amount.lean`: `paymentUpper_is_exact`, `resolve_selects_the_funded_full_balance`	`X/Consumers/OriginalPayment.lean`: full six-unit payment and victim refusal retaining the original debt
19	`X/Network.lean`: `opens_network_at_the_original_resident`; `X/Renewal.lean`; `R/History.lean`	`X/Consumers/{OriginalNetwork,OriginalRevision/Repeated,OriginalShortage}.lean`
20	`R/Decision/Pending.lean`: `pending_completion_uses_the_posted_receipt`; `R/Decision/Observation.lean`: `every_finite_duration_reader_has_a_hidden_protected_change`, `no_finite_duration_reader_decides_the_paid_scope`	`R/Decision/Consumers/{Original,Pending}.lean`: diagonal undecidability and same-receipt completion
21	`R/Privacy/History.lean`: `every_unpaid_prefix_keeps_the_transcript_and_language`, `a_new_distinction_requires_the_same_actual_payment_on_both_sides`	The actual common reserve from PE19 and both histories' actual continuations
22	`E/Protection.lean`: `protected_frontier_exact`, `shortest_protected_exact`, `longest_protected_exact`; `C/Emergency/Horizon.lean`: `no_reset_on_any_original_successor`, `no_authority_after_a_crossing`; `E/{Calendar,Envelope,Solver,Publication,Lifecycle}.lean`	`E/Consumers.lean`: clocks $1, 3, 4$, two actual review payments, missing exit, and post-crossing refusal
23	`X/Scheduled/{Due,Motion,History}.lean`: `writeDue`, `every_round_consumes_the_original_successor`, `the_unsettled_clock_is_cofinal`	`X/Consumers/OriginalDue/Settlement.lean`
24	`O/Operational/Joint/Search.lean`: `search_is_complete_at_every_width`, `checked_prefixes_cover_every_finite_joint_recipe`; `R/Rebase/Completeness.lean`: `available_joint_production_forces_the_first_write`	`T/Deployment/Replacement.lean`: width-four first write and width-one due-payment comparison
25	`R/Decision/Deferred/Transport.lean`: `Deferred.completion_along_reads_the_same_original_scope`, `Deferred.covered_paid_scope_does_not_write_twice`; `R/Rebase/Privacy.lean`	`X/Consumers/OriginalRevision/{Rebase,Mixed}.lean`; `T/Deployment/Continuation/Waiting.lean`
26	`Lean/SaturationMonoid/Tracks/LivingLawTerminalPeaceInstance.lean`; `T/Deployment/{Facade,Ingress,Continuation/Waiting}.lean`	`Lean/scripts/{living_law_runtime,terminal_peace_worker}.py`: batch clocks $5 \to 13$ (13 checks) vs persistent `Ingress` U8 write at 10 and four-host head at 12; original-ID readback and same-parent transactions

PE1–PE33 public declarations and complete types. public claim map · reproduction guide

Supporting proofs at Homework `3a90cba5ac`. PE3 Payer · ReceiptAccount · Outstanding · Guard. PE6 Root.

Appendix I. Certified Histories, Handover, and Interrupted Attempts

I.1. Certified objects and signing obligations

Let P be the type of ledger heads and M the type of verifier identities. Each head $p \in P$ obtains a four-member roster $O(p) \subset M$ from its already certified history, instantiating the $n = 3f + 1 = 4$ ($f = 1$), quorum-size $2f + 1 = 3$ intersection geometry against equivocation and seal violation at every parent head. A successor q contains the original request identifier, the literal input, the actual payment at that occurrence, the resulting readout, and the next roster. Every attestation binds the same parent head p and candidate successor q.

Write $V(m, p, q)$ for member m's durable decision vote for this successor, and $R(m, p, q)$ for its attestation that it replayed the same payment from the original source. Define certification by

$$\begin{aligned} \mathrm{Cert}(p, q) \iff & [\exists Q \subseteq O(p),\ |Q| \geq 3,\ \forall m \in Q,\ V(m, p, q)] \\ & \wedge [O(q) \neq O(p) \Rightarrow \forall m \in O(q),\ R(m, p, q)]\,. \end{aligned} \tag{I.1}$$

Each handover connects an outgoing three-member decision quorum to the incoming roster's complete replay of that occurrence. The roster takes effect through this edge, and the next write is verified under the new roster.

Ordinary certification uses the following durable signing constraint. At each parent p, at most one member belongs to the fault set B_p; every other member may vote for only one successor at that parent. The fault set may change with the parent. When withdrawal and reassignment are allowed, the constraint becomes: if a member that observes the constraint has voted for two distinct candidates, at least one candidate must have a valid exclusion. We derive sound exclusion from role-specific seals below, then prove that this weaker signing constraint still preserves a unique history.

I.2. Quorum intersection through arbitrarily many handovers

Lemma I.1 (An intersection outside the fault set). Let $|O| = 4$, $|B| \leq 1$, and $W, U \subseteq O$, with $|W|, |U| \geq 3$. Then $W \cap U$ contains a member outside B.

Proof. Since $W \cup U \subseteq O$,

$$|W \cap U| = |W| + |U| - |W \cup U| \geq 3 + 3 - 4 = 2.$$

A fault set of size at most one cannot contain this intersection. Choose an intersecting member outside it. □

Theorem I.2 (A unique prefix order for certified histories). Suppose every parent head satisfies the four-member, single-fault, and durable signing conditions of Lemma I.1. Then:

1. Any two certified successors of the same parent are equal.
2. Any two finite certified histories from the same root are prefix comparable.
3. Any two infinite certified continuations from the same root agree at every write index.

Successive rosters may be completely disjoint.

Proof. Each certified edge provides an outgoing decision quorum. Lemma I.1 finds a member outside the fault set who has voted for both edges. Its durable signing constraint forces the successors to coincide, proving the first assertion. For finite histories, induct on their length: the empty history is a prefix of every history; two nonempty histories have the same first successor, so the induction hypothesis applies at that successor. For infinite histories, begin at their common root and repeatedly apply successor uniqueness. The argument at each step uses only the roster of that parent, so no overlap between successive rosters is required. □

Lemma I.1 also applies to three fresh read responses and the three members that durably stored a write. Two read responses do not give the same guarantee. Take $O = \{0, 1, 2, 3\}$, durable writers $\{0, 1, 2\}$, readers $\{2, 3\}$, and sole faulty member 2. The two sets intersect only at the faulty identity. This is the exact counterexample underlying the three-read threshold for cold recovery.

I.3. An exact completion criterion after an interrupted handover

Fix one parent head, one epoch, and one handover candidate. Let O be the outgoing roster and N the proposed incoming roster, each with four members. Let $D \subseteq O$ record outgoing **no-decision seals**, and let $S \subseteq N$ record incoming **no-replay seals**. A seal carries two durable obligations: the member has not previously signed that role and will not sign that role subsequently.

For an outgoing triple $Q \subseteq O$, define

$$F(Q) = (Q \cap D) \cup (N \cap S). \tag{I.2}$$

Call a pair (B, C) of finite sets a **compatible completion** under (D, S) if the outgoing decision set $C \subseteq O$ satisfies $|C| \geq 3$, the fault set B satisfies $|B \cap O| \leq 1$ and $|B \cap N| \leq 1$, and $(C \cap D) \cup (N \cap S) \subseteq B$. If the votes of Q and the replays of all of N can jointly form a certificate, every member of $Q \cap D$ has both voted and sealed against voting, while every member of $N \cap S$ has both replayed and sealed against replay. These identities must therefore belong to the fault set. An overlapping identity is counted once in the union; each roster checks its own fault budget.

Theorem I.3 (Exact completion criterion). A compatible completion (B, C) respecting the fault budget of at most one member in each roster exists under (D, S) if and only if

$$\exists Q \subseteq O, \quad |Q| = 3, \qquad |F(Q) \cap O| \leq 1, \qquad |F(Q) \cap N| \leq 1. \tag{I.3}$$

When $|O| = 4$, exactly four outgoing triples need to be examined.

Proof. If (I.3) holds, take the fault set to be exactly $F(Q)$. The outgoing triple Q supplies the decision votes and the incoming roster N supplies the replays. Every identity whose role conflicts

with a seal is already included in this fault set, and both roster budgets are satisfied. This constructs a combination of certifying roles compatible with the evidence.

Conversely, suppose a compatible fault set B and outgoing signer set $C \subseteq O$ exist, where $|C| \geq 3$ and $|B \cap O|, |B \cap N| \leq 1$. Choose a triple $Q \subseteq C$. Every identity in $Q \cap D$ or $N \cap S$ must belong to B, so $F(Q) \subseteq B$. Intersecting with either roster preserves the cardinality bound of one, yielding (I.3). □

Compatibility answers a specific question: does the current evidence still permit the original candidate to obtain complete certification? The execution returns the witness $Q, N, F(Q)$, keeping the missing evidence explicit. Only when all four triples fail (I.3) does the evidence exclude the original candidate.

Corollary I.4 (A certified candidate cannot be validly excluded). If the original candidate is already certified and identities that contradict role-specific seals respect the fault budget in each roster, then the candidate continues to satisfy (I.3).

Proof. Use the certificate's actual outgoing signer set and the actual fault set. Every contradiction between an attestation and a seal lies in that fault set. The converse argument in Theorem I.3 supplies a compatible triple. □

Corollary I.5 (Three seals exclude; two can remain compatible). Three outgoing no-decision seals in a four-member roster suffice to exclude the original handover. Two such seals can remain compatible.

Proof. Every outgoing decision triple intersects a three-member seal set in at least two members. Thus $F(Q) \cap O$ contains at least two identities, exceeding the outgoing fault budget. For the counterexample, take

$$O = N = \{0, 1, 2, 3\}, \quad D = \{0, 1\}, \quad S = \emptyset, \quad Q = \{0, 2, 3\}.$$

Here $F(Q) = \{0\}$; the fault count is one in both rosters, so (I.3) holds. □

This counterexample shows that two outgoing seals alone cannot support a universal exclusion rule. Each actual candidate is still checked by substituting all available outgoing and incoming role-specific seals into (I.3).

I.4. Checking the maximum obtainable evidence first

A seal permanently closes the corresponding signing role, so the order in which seals are issued is itself part of the execution.

Suppose the current responders can supply at most $D_{\max}, S_{\max}$. If the original candidate satisfies (I.3) even for these maximum seal sets, then it also satisfies the criterion for every $D \subseteq D_{\max}$, $S \subseteq S_{\max}$. For the same witnessing triple,

$$(Q \cap D) \cup (N \cap S) \subseteq (Q \cap D_{\max}) \cup (N \cap S_{\max}),$$

so both cardinality constraints can only weaken. The execution therefore stops before requesting irrevocable seals, returns the named evidence gap, and preserves the route for completing the original certificate under the original request identifier.

An ordered handover first durably retains the original intent at at least three outgoing nodes, then obtains all four incoming replays. Outgoing decision votes must consume that complete replay evidence.

Corollary I.6 (Deadlock-free disposal before replay completion). Suppose $|O| = 4$, $D \subseteq O$, and casting an outgoing decision vote requires complete incoming replay. If the handover is interrupted before incoming replay is complete, then even if all four nominated incoming members stay unreachable (no incoming responses), responses from any set $R_{\text{out}} \subseteq O$ of at least three reachable outgoing members—each returning either an actual decision vote or a no-decision seal in D—necessarily refute (I.3) and validly exclude the attempt.

Proof. Every $m \in R_{\text{out}}$ either has cast a decision vote or satisfies $m \in D$. Because incoming replay is incomplete, no outgoing decision vote can exist, forcing $R_{\text{out}} \subseteq D$ and hence $|D| \geq |R_{\text{out}}| \geq 3$. Corollary I.5 immediately excludes the candidate. □

The execution thus has three outcomes:

Available evidence	Actual successor
The original certificate is complete	Complete the original request identifier and recover the same original payment
A compatible combination remains	Return the outgoing triple, required faulty identities, and missing evidence; retain the original signing route
All four triples are excluded	Under the original caller's authorization, submit an actual due payment at the same parent head and record withdrawal or reassignment together with all exclusion evidence

With repeated reassignment, each excluded candidate's request identifier, candidate head, original authorization, and proposed roster are retained in the final paid row. A later lookup of an original identifier returns its own disposition and the actual successor.

I.5. Unique histories with valid exclusion

Write $E(p, q)$ when candidate q has a valid exclusion at parent p. Assume

$$\text{Cert}(p, q) \Rightarrow \neg E(p, q), \tag{I.4}$$

and that at each parent at most one member of the four-member roster violates the signing constraint

$$V(m, p, q_1) \wedge V(m, p, q_2) \Rightarrow q_1 = q_2 \ \vee \ E(p, q_1) \ \vee \ E(p, q_2). \tag{I.5}$$

Corollary I.4 supplies (I.4). Nodes enforce (I.5) by verifying the candidate's exclusion chain within a durable transaction.

Theorem I.7 (Arbitrary dispositions preserve the common-root history). Under these conditions, any two finite certified histories from the same root are prefix comparable, and any two infinite certified continuations from that root agree at every index. The conclusion covers arbitrarily many handovers, withdrawals, and reassignments.

Proof. Define a surviving vote by

$$V^{\circ}(m,p,q) := V(m,p,q) \wedge \neg E(p,q).$$

By (I.4), every vote on an original certified edge becomes a surviving vote, so every original certified history is retained. If a member that observes the signing constraint has surviving votes for q_1, q_2, the last two alternatives in (I.5) contradict the survival conditions. Hence $q_1 = q_2$. Surviving votes therefore satisfy the strict durable signing condition of Theorem I.2. That theorem gives finite prefix comparability and infinite pointwise agreement. □

The proof retains every already certified edge and releases only candidates with valid exclusions. Withdrawal allows a verifier to take part in the actual payment at the original parent without changing any established history.

I.6. The certified execution of the original six-unit ledger

Three frozen receipts successively consume the same source, `terminal-peace/original-six`. In every paid successor below, the original debt address is $(2, 0, (\text{false}, \text{true}, \text{false}))$, and the reparation balance is zero. The clock is generated by actual payments in the source execution; request identifiers and visit cursors are retained separately.

Source clock	Recorded execution	Roster after completion
12	Shared original head before four-host admission	$abcd$
13	A three-vote wait is followed by all four native replays, completing the first admitted write	$abcd$
14	Three nodes attest the next actual renewal payment	$abcd$
15	The returning node replays the exact original prefix before attesting	$abcd$
16	The installed entry point makes an actual payment; the original caller and four nodes seal this prefix	$abcd$
17	After loss of the primary journal and local intent, the original payment is completed under its original identifier; two fresh read responses had named b, c as missing	$abcd$
18	Paid handover $abcd \to abce$; absence of e halts at incoming replay, and completion follows when its evidence arrives	$abce$
19	After rollback of the primary to clock 16, the latest original ledger is reconstructed from certified history	$abce$
20	The installed entry point pays; the three remote hosts b, c, e alone recover the original head and request identifier	$abce$
21	An actual payment activates the ordered entry: replay precedes decision voting	$abce$
22	Remote seals recover the original paid withdrawal; late replay cannot cross the outgoing seals	$abce$
23	Valid exclusion is followed by paid reassignment, completing $abce \to abcf$	$abcf$
24	Recovery of the third actual decision vote completes the already selected original identifier and the handover $abcf \to abcg$	$abcg$
25	Outgoing seals from b, c, g exclude a candidate with one stored vote; that voter verifies exclusion and signs the paid withdrawal	$abcg$
26	The installed entry point pays the next due; three remote hosts recover the current-roster ledger	$abcg$

Source clock	Recorded execution	Roster after completion
27	Two stored decision votes still admit a compatible combination; the original route is retained and the same identifier completes $abcg \to abcf$	$abcf$
28	The next due is paid, and three remote hosts recover the same final head	$abcf$

The actual certified rows at clocks 21–28 are listed below. An empty incoming column means that the payment retains the current roster. Clock 24 has exactly three outgoing decision votes; the other records retain all four votes received at those occurrences.

Source clock	Actual request identifier	Outgoing decisions	Incoming replays
21	`ordered-protocol-21`	a, b, c, e	—
22	`withdraw-f-22`	a, b, c, e	—
23	`reassign-to-f-23`	a, b, c, e	a, b, c, f
24	`hidden-selected-g-24`	a, b, c	a, b, c, g
25	`recovered-capacity-25`	a, b, c, g	—
26	`ordered-peace-continues-26`	a, b, c, g	—
27	`two-outgoing-live-27`	a, b, c, g	a, b, c, f
28	`guarded-ordinary-28`	a, b, c, f	—

The thresholds have distinct roles. Initial four-host admission and bootstrap sealing require four attestations; subsequent ordinary writes and fresh reads of the current roster require at least three. A handover certificate requires at least three outgoing decision votes and all four incoming replays. Normal completion also durably stores the complete certificate at at least three nodes in each roster. At clock 18, the incoming identity e uses the retired identity d's original host with a separate new key; physical hosts and changes of logical identity are recorded separately. An independent local regression exercises the completely disjoint handover $abcd \to efgh$ under the same contract.

I.7. Claims, proofs, and direct acceptance evidence

This appendix binds theorems, process regressions, and physical-node executions separately. The Lean theorems quantify over arbitrary identities, heads, and admissible certified histories. Local regressions test fault schedules with either a deterministic native interface or the actual Lean entry point. The four-host receipts preserve the execution of the same original six-unit source described above.

In the table, $D =$ `Lean/SaturationMonoid/ProcessGame/Society/Institutions/Conflict/Constitution/TerminalPeace/Deployment/` and $P =$ `Lean/scripts/terminal_peace_replication/`.

PE	Fixed Homework revision	Original producer or execution entry	Direct consumer, counterexample, and acceptance
PE27	`3a90cba5ac7427737ee2fe630612dfed43e45da1`, original increment `c882831f9a`	`terminal_peace_anchor.py`, `terminal_peace_submit.py`, `terminal_peace_worker.py`	`terminal-peace-admission-check.py`; four-host `acceptance.json`, clocks 12–16, retention of the original intent when votes are missing, and rejection of an older parent at all four nodes
PE28	`b5b9c5f86cab0bb212e306ab89db4bf5b177a177`	*P*`replica.py::Replica.prepare/status/store`, `quorum.py::Quorum.latest`, `operator.py::Primary.recover/install`	Lost journal, stale challenge, original identifier, rollback, and partial archive in `terminal-peace-replication-check.py`; four-host `recovery-acceptance.json`
PE29	As PE28	*P*`protocol.py::Timeline.candidate/verify_votes/apply`, `Quorum.choose/store`	The same native `due` and the clock-18 handover; the local $abcd \to efgh$ consumer
PE30	As PE28	*D*`Recovery/Quorum.lean`: `honest_overlap`, `unique_successor`, `original_histories_are_prefix_comparable`, `cofinal_histories_agree`	`two_reads_can_miss_an_honest_durable_verifier`; the quorum contract for cold recovery and handover
PE31	`cf880bb282f0ff28d6e9f9fc890341baaa8b0eb0`	*D*`Recovery/HandoffDisposition.lean`: `possible_iff_completion`, `possible_with_maximal_seals`, `selected_stays_possible`, `three_outgoing_seals_exclude`	`two_outgoing_seals_can_leave_a_completion`, `interrupted_before_decision_can_be_disposed`; *P*`rounds.py::possible_completion`, `Quorum.resolve`
PE32	As PE31	`dispositions_preserve_original_prefix_order` and `dispositions_preserve_every_cofinal_cut` in the same file	*P*`Timeline.exclusion/resolved/apply`, `Replica.stage/ready/seal/prepare`, `Primary.resolve/recover/read_result`
PE33	As PE31	The original execution above and the `terminal_peace_online.py` entry point	Repeated reassignment, late replay, hidden selection, and preservation of two live votes in `terminal-peace-handoff-resolution-check.py`; four-host `handoff-resolution-acceptance.json`, clocks 21–28

The three frozen receipts are linked below with their original SHA-256 digests.

Receipt	SHA-256
acceptance.json	c087bd309a4e92b62ac3d88a17ec219c710e9fee929532a8a861dca26a6d968d
recovery-acceptance.json	26522f53e07161b9bd8504c609292295168b90f6ba7927ad77c2e22cbc2847ad
handoff-resolution-acceptance.json	7fe676f6d8dd085b959c92aa5ca151b7e2a9478378306a80a44d82924733c56b

Each later receipt retains the preceding receipt's digest. The original kernel acceptance records `propext`, `Classical.choice`, and `Quot.sound` for the five ordinary recovery declarations and the eight disposition declarations. The original source SHA-256 digests recorded in the frozen receipts for `Quorum.lean` and `HandoffDisposition.lean` are `09dc0782ab9361c0be28f1b1fd83ab07bf6df3773dbe80a17485c3044a2f3e34` and `7d3ba9fbd16aa63c3ce3003ae5cf4132f76380835dd84cfe26dcb4bce83ebfee` (in the public selection, `HandoffDisposition.lean` rewrites only its one-line versioned `import`, and `python3 tools/enforceable_peace.py source` recovers the exact original digest byte for byte).

Offline acceptance verifies 95 public-key signatures covering initial admission, bootstrap sealing, the clock-18 handover, and clocks 21–28. It recomputes the eight new heads, checks their successive parents, original debt address, zero balance, and clock continuity, and enumerates the four outgoing triples at each of the three dispositions. The identities of the physical-node executions are fixed by the three original receipts.

The public map for PE1–PE33 identifies the fixed execution programs and direct consumers. The offline replay uses the original certified journal and verifies the same Lean continuation through clock 28.